

Structural Dynamics and Contagion Mechanisms in Event-Driven ETF Analysis

Research Plan

1 Introduction and Research Motivation

Previous research shows that ETF ownership increases return comovement, volatility and liquidity commonality among underlying stocks. However, most studies focus on aggregate comovement patterns or on shocks originating in ETF trading activity, rather than on firm-specific corporate events.

This project aims to study whether firm-specific corporate events affecting large ETF constituents generate spillover effects within the ETF via propagation to other stocks in the same ETF.

The goal is to understand the mechanism behind any observed spillovers. Specifically, the project will test whether these effects are driven by:

- Ownership-related synchronization
- Mechanical trading channels (ETF arbitrage and flows)
- Informational spillovers across related firms

2 Literature Review

2.1 ETF Arbitrage and Shock Transmission

The literature studies how ETF arbitrage can transmit shocks between ETFs and their underlying securities.

In *“ETFs, Arbitrage, and Contagion”* (Ben-David, Franzoni, and Moussawi 2012), the authors show that when arbitrage is limited, mispricing between ETF prices and their Net Asset Value can transmit non-fundamental shocks to underlying stocks. Their results suggest that ETF arbitrage may amplify volatility and create temporary price pressure in constituent stocks.

In *“Do ETFs Increase Volatility?”* (Ben-David, Franzoni, and Moussawi 2018), the same authors find that higher ETF ownership is associated with higher volatility in underlying stocks. So ETFs could introduce an additional layer of liquidity-driven shocks.

2.2 ETF Ownership and Return Comovement

Another line of research studies how ETF ownership affects return comovement.

In *“Exchange-Traded Funds and Asset Return Correlations”* (Da and Shive 2018), the authors document that ETF activity is associated with higher return correlations among constituent stocks. Thus, ETF trading can synchronize price movements across firms.

Similarly, *“Do ETFs Increase the Comovements of Their Underlying Assets?”* (Marta and Riva 2025) shows that ETF ownership and the creation or redemption mechanism are linked to stronger comovement in both returns and liquidity.

Related evidence for European markets is provided in *“Effects of ETF Ownership on the Volatility and Return Comovement of European Stocks”* (Sakss 2019), which finds that greater ETF presence is associated with stronger cross-sectional comovement.

From an informational perspective, *“Is There a Dark Side to Exchange Traded Funds? An Information Perspective”* (Israeli, Lee, and Sridharan 2017) argues that ETF ownership may reduce the firm-specific component of price discovery. As ETF trading becomes more important, stock prices may reflect more systematic movements and less firm-level information.

2.3 Liquidity Commonality and Contagion

In *“Sectoral Integration, Comovement and Contagion”* (Cho, Hyde, and Nguyen 2015), the authors document how shocks propagate within economically related groups of assets. In the ETF context, studies analyzing liquidity commonality suggest that ETF trading may increase synchronization in liquidity conditions across constituent stocks. Therefore, liquidity shocks affecting one stock may indirectly affect others through shared trading mechanisms.

2.4 ETF Flows and Price Pressure

In *“ETF Flows and Underlying Stock Returns: The True Cost of NAV-Based Trading”* (Osterhoff and Overkott 2016), the authors show that ETF creations and redemptions can generate measurable abnormal returns in underlying stocks, particularly through the in-kind creation and redemption mechanism. This suggests that ETF flows can create mechanical price pressure.

3 Research Gap

As seen above, the literature shows that:

- ETF arbitrage can transmit shocks from ETF markets to underlying assets (Ben-David, Franzoni, and Moussawi 2012).
- ETF ownership is associated with higher volatility (Ben-David, Franzoni, and Mousawi 2018).
- ETF activity increases return comovement (Da and Shive 2018; Marta and Riva 2025).

- ETF trading may reduce firm-specific price discovery (Israeli, Lee, and Sridharan 2017).
- ETF flows can generate price pressure in underlying stocks (Osterhoff and Overkott 2016).

That said, there is limited evidence on whether firm-specific corporate events affecting large ETF constituents generate spillovers to other stocks within the same ETF.

In addition, it remains unclear through which mechanism such spillovers operate: mechanical trading channels (flows and arbitrage), ownership-induced synchronization, or informational spillovers.

This study will address this gap by focusing on event-driven spillovers within ETFs.

4 Research Questions

This set of questions defines the research plan:

1. Do firm-specific shocks to major ETF constituents generate abnormal returns in other constituent stocks?
2. Is the propagation effect stronger for negative shocks than for positive shocks?
3. Does liquidity amplify the transmission mechanism?
4. Does the weight of the shocked stock within the ETF affect spillover magnitude?
5. Are spillovers amplified on days with larger ETF flows or trading activity?

5 Hypotheses

H1 – Intra-ETF Propagation: Firm-specific shocks to large-weight constituents generate statistically significant abnormal returns in other constituent stocks of the same ETF.

H2 – Liquidity Channel: Spillovers are stronger among less liquid constituent stocks.

H3 – Weight Channel: The magnitude of spillovers increases with the weight of the shocked stock in the ETF.

H4 – Flow Amplification: Spillovers are stronger on days with higher ETF trading activity.

H5 – Asymmetry: Negative shocks generate stronger spillover effects than positive shocks.

References

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